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DETAIL SPECIFICATION

TORSION BAR, SUSPENSION

This specification is approved for use by the U.S. Army Tank Automotive Research, Development and Engineering Center (TARDEC), Research, Development and Engineering Command (RDECOM), Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a military combat vehicle suspension torsion bar.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4 or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Government documents, drawings, and publications. The following Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract (see 6.2).

Comments, suggestions, or questions on this document should be addressed to U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to DAMI_STANDARDIZATION@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil.

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U.S. ARMY TACOM DRAWINGS

12369003	-	Steel Ext. Surfaces, Green 383
12292010	-	Torsion Bar, Suspension
10929033	-	Gage, Special Involute Go Composite Ring
10929035	-	Gage, Involute Not Go Sector Ring

(Copies of these documents are available from DAMI_STANDARDIZATION@conus.army.mil or U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

ASTM INTERNATIONAL

ASTM A604	-	Method for Macroetch Testing of Consumable Electrode Remelted Steel Bars and Billets (DoD Adopted)
ASTM E1444	-	Standard Practice for Magnetic Particle Examination

(Copies of these documents are available from www.astm.org or ASTM International, P.O. Box C700, West Conshohocken, PA 19428-2959.)

SAE INTERNATIONAL

SAE AMS 2300	-	Steel Cleanliness, Premium Aircraft - Quality Magnetic Particle Inspection Procedure (DoD Adopted)
SAE AMS 6257	-	Steel Bars, Forgings, and Tubings (DoD Adopted)
SAE AMS-STD-2154	-	Inspection, Ultrasonic, Wrought Metals, Process for

(Copies of these documents are available from www.sae.org or SAE Customer Service, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.4 (see 6.3).

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3.2 Design and construction.

3.2.1 Materials. The contractor shall select the material, but the material shall be capable of meeting the operational requirements specified herein. It is recommended, but not mandatory, that steel alloy 300M, class 3, condition optional, as specified in SAE AMS 6257 be used.

3.2.1.1 Billet material. The billet material shall meet both of the following requirements:

a. The billet material shall have a sufficient internal cleanliness as determined by testing in accordance with 4.2a.
b. The billet material shall have a sufficient surface cleanliness as determined by testing in accordance with 4.2b.

3.2.1.2 Bar stock. The bar stock shall meet both of the following requirements:

a. The finished bar stock (before hot upsetting) shall have a sufficient internal cleanliness as determined by testing in accordance with 4.2c.
b. The finished bar stock shall be as follows:
    1. The finished bar stock (before hot upsetting) shall have a sufficient surface cleanliness as determined by testing in accordance with 4.2d. No laps or seams shall be permitted, or,
    2. The finished bar stock shall have a sufficient internal cleanliness as determined by testing in accordance with 4.2e.

3.2.2 Processing.

3.2.2.1 Heat treatment. The torsion bar shall be quenched and double tempered to a through hardness of Rockwell C51 minimum. This hardness is required for the bar to be compatible with the design requirements of interfacing components (see 4.6.1).

3.2.2.1.1 Decarburization and retained austenite. Microscopic examination of a prepared cross section of the torsion bar following final heat treatment at a magnification of 250 diameters shall show no evidence of retained austenite or partial decarburization beyond a depth of 0.003 inches from the surface. Partial decarburization between the surface and a depth of 0.003 inches shall not result in a hardness reduction of more than 3 points Rockwell C as determined by microhardness traverse comparison between that area and the core hardness (see 4.6.1).

3.2.2.2 Surface indications. Magnetic particle inspection of the finished torsion bar (after final heat treatment) shall disclose no indications in excess of the requirements of SAE AMS 2300 and SAE AMS 6257. Laps, inclusions, and seams shall not be permitted. Nicks and scratches less than four inches long shall be removed by blending the surface to a depth of not greater than 0.010 inch. After removal of scratches, the torsion bar shall still meet dimensional requirements of Drawing 12292010 (see 4.6.2 and 4.6.3).

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3.2.2.3 Surface finish. The surface texture of a torsion bar shall be as specified on Drawing 12292010 (see 4.6.3).

3.2.2.4 Preset. The torsion bar shall be preset to the requirements of Drawing 12292010 by twisting one end of the bar in the direction indicated a minimum of three times while restraining the opposite end. The bar shall be held stationary at full windup for not less than 10 seconds during each preset cycle. The last twist shall result in a change of set angle of not greater than 1.0 degree from the preceding twist. The degrees of permanent set shall be not greater than the amount specified on Drawing 12292010 (see 4.6.4).

3.2.2.5 Straightness. Straightness of the torsion bar body shall be controlled as specified on Drawing 12292010. Stock or bars may be straightened at any time during the machining sequence as required to meet the drawing requirements for as-machined straightness prior to any cold working. Note that the drawing straightness control for as-machined torsion bars is tighter than the final straightness requirement. Bars shall be straightened at a temperature below the tempering temperature. Care shall be exercised to avoid damage to the surface of the bar during the straightening operation.

3.2.3 Surface protection.

3.2.3.1 Cleaning. Prior to the application of any protective coating, bars shall be cleaned in accordance with Drawing 12369003, method I or method II, with no film thickness specified.

3.2.3.2 Priming. The body of the torsion bar shall be primed with primer coating conforming to Drawing 12369003 with no film thickness specified.

3.2.3.3 Protective covering. A protective covering shall be applied to the body of the bar in the area shown on Drawing 12292010. This covering shall prevent damage from handling or shipping such as scratches, nicks or abrasions (see 6.6).

3.2.4 Dimensions. Dimensions and tolerances of the torsion bar shall be as specified on Drawing 12292010.

3.2.5 Endurance. The torsion bar shall demonstrate a minimum life of 45,000 cycles and an average life of 80,000 cycles with an angular creep of not greater than 4 degrees and torsional deflection as specified in table I (see 4.6.5).

TABLE I. Endurance.

Part number 12292010	Torsional deflection (degrees)
-1, -2	12-71
-7, -8	12-69

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3.3 Product marking. The part identification number, mill heat marking, preset arrow and manufacturer's identification shall be applied to the ends of the torsion bar as specified on Drawing 12292010.

3.4 Workmanship. The torsion bar shall be free of defects such as scratches, nicks, tool marks, burrs, sharp edges, scratched finish, peeled or blistered primer, or improperly applied protective covering (see 4.6.3).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Conformance inspection (see 4.5).

4.2 Materials discontinuities and flaws. Materials shall be free of discontinuities and flaws that will affect torsion bar function and, as a minimum, the materials shall pass the following tests:

a. To determine conformance with 3.2.1.1a, all billet materials shall pass the ultrasonic test requirements of SAE AMS-STD-2154, type I or II, class A.
b. To determine conformance with 3.2.1.1b, all billet materials shall be macroetched in accordance with ASTM A604 at each crop location. The macrostructures shall be equal to or better than severity A for class 1 or 2 and severity B for class 3 and 4.
c. To determine conformance to 3.2.1.2a, all finished bar stock shall pass the ultrasonic test requirements of SAE AMS-STD-2154, type I or II, class AA.
d. To determine conformance with 3.2.1.2.b.1, all finished bar stock shall pass 100 percent magnetic particle inspection using the fluorescent wet continuous method in accordance with ASTM E1444, except waive the frequency and severity requirements as specified in SAE AMS 2300.
e. To determine conformance to 3.2.1.2.b.2, all finished bar stock shall pass the ultrasonic test requirements of SAE AMS-STD-2154, type I or II by the Shear Wave (Angle Beam) Technique. Equipment calibration shall be performed using a cylindrical specimen of the same diameter and surface finish as the finished product containing a groove not greater than 0.004 inches deep by 4 inches long, oriented in the longitudinal direction of the bar. Actual dimensions of the notch shall be measured and marked on the cylindrical specimen.

4.2.1 Billet material properties. If required by the Government (see 6.2), the contractor shall provide a sample of the billet material from each mill heat crop location for verification of material properties. The sample shall be produced using the supplier's standard manufacturing practices.

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4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following standard (room) ambient conditions:

a. Temperature: 77±18 degrees Fahrenheit (°F)
b. Relative humidity: Uncontrolled room ambient
c. Atmospheric pressure: Site pressure

4.4 First article inspection.

4.4.1 Sample size. Six torsion bars of the 12292010-1 configuration shall be subjected to first article inspection.

4.4.2 Inspection routine. The first article samples shall be subjected to the group A, B, C and D inspections below, except that the 100,000 cycle minimum of 4.5.2 shall not apply.

4.4.3 Failure. Failure to pass group A, B and C inspections shall be cause for refusal to grant first article approval. Each torsion bar subjected to group D inspection shall demonstrate a minimum life of 45,000 cycles. Average life calculation of 80,000 cycles will be based on the total number of endurance cycles accumulated during testing divided by the total number of torsion bars tested. If a torsion bar exhibits endurance life of 200,000 cycles or more prior to failure, its life shall be taken as 200,000 cycles for the average life calculation. Failure to meet group D minimum life or average life requirements shall be cause for refusal to grant first article approval.

4.4.4 Disposition of samples. Samples which have been subjected to first article inspection shall not be delivered as new equipment under the contract.

4.5 Conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A, B and C inspections.

4.5.1.1 Inspection lot. Unless otherwise noted, an inspection lot shall consist of all torsion bars produced under essentially the same conditions and offered for inspection at one time. The lot may include a mixture of the torsion bar configurations specified on Drawing 12292010. Sufficient quantities of 12292010-1 configuration shall be fabricated to verify conformance to 4.4.1.

4.5.1.2 Group A inspection. Group A inspection shall consist of the inspections specified in table II.

4.5.1.2.1 Group A sampling. 100 percent inspection shall apply unless an approved statistical process control (SPC) plan is used (see 6.4.2).

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4.5.1.2.2 Group A failure. The supplier may rework a rejected torsion bar to correct the defect(s) and resubmit it for inspection. Rework shall be within the confines of drawing and specification requirements.

4.5.1.3 Group B inspection. Group B inspection shall consist of the inspections specified in table III.

TABLE II. Group A inspection.

Inspection	Requirement	Inspection method
Surface indications	3.2.2.2	4.6.2
Surface finish	3.2.2.3	4.6.3
Workmanship	3.4	Visual
Spline major diameter	3.2.4	Go Ring Gage and Not Go Ring Gage
External involute spline	3.2.4	Special Involute Go Composite Ring Gage 10929033 and Involute Not Go Sector Ring Gage 10929035

4.5.1.3.1 Group B sampling. Frequency of inspection and/or any special arrangements for establishing the frequency of inspection shall be as specified in the contract or purchase order (see 6.2). Recommended requirements for a sampling plan are shown in 6.4.

4.5.1.3.2 Group B failure. The supplier may rework a rejected torsion bar to correct the defect(s) and resubmit it for inspection. Rework shall be within the confines of drawing and specification requirements.

4.5.1.4 Group C inspection. Group C inspection shall consist of the process inspections specified in table IV.

4.5.1.4.1 Group C sampling.

4.5.1.4.1.1 Heat treatment. A minimum of one sample torsion bar or control specimen shall be selected from each heat treatment lot.

4.5.1.4.1.2 Preset and protective coating. Unless an approved SPC plan is used (see 6.4.2), a minimum of three torsion bars shall be selected at random during each work shift for inspection of preset and protective coating.

4.5.1.4.2 Group C failure. Failure of a torsion bar or control specimen (see 4.6.1.1) to meet the specified requirements in table IV shall be cause for rejection of the torsion bars represented. Acceptance of production lots subsequent to failure shall not resume until corrective action has been taken and three consecutive lots pass group C inspection.

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4.5.1.4.3 Group C rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective torsion bars and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

TABLE III. Classification of defects.

Category	Defect	Method of examination
Major:		
101	Dimensions affecting interchangeability (see 3.2.4).	SIE 1/
102	True position of spline pitch diameters and angle between removed spline teeth (see 3.2.4).	Surface plate, vee blocks and special involute Go Composite Ring Gage10929033
103	Spline fillet root radius (see 3.2.4).	Radius gage and magnifying glass
104	Surface hardness, HRC 51 minimum (measured on each end face or removed tooth area) (see 3.2.2.2).	Hardness tester
Minor:		
201	Dimensions not affecting interchangeability (see 3.2.4).	SIE
202	Preset direction arrow compatible with part number (see 3.2.2.4).	Visual
203	Body straightness (see 3.2.2.5).	Surface plate, vee blocks, dial indicator and stand
204	Protective covering not properly applied (see 3.2.3.3).	Visual
205	Identification, mill heat and manufacturer markings (see 3.3).	Visual

1/ SIE = Standard Inspection Equipment.

TABLE IV. Group C inspection.

Inspection	Requirement	Inspection method
Heat treatment	3.2.2.1, 3.2.2.1.1	4.6.1 and hardness tester
Preset	3.2.2.4	4.6.4
Surface protection	3.2.3	Visual

4.5.2 Group D inspection (control testing). Group D inspection shall consist of the inspection specified in table V. Group D inspection shall be made on sample torsion bars from each configuration in the lot that have been subjected to and passed groups A, B, and C inspection. The acceptance criteria for group D inspection shall be as specified in 3.2.5 except

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that the torsion bar shall demonstrate a minimum life of 100,000 cycles. Delivery of torsion bars which have passed group A, B and C inspection shall not be shipped until control test is successfully completed (for exception see 6.2).

4.5.2.1 Group D sampling. Unless otherwise specified (see 6.2), torsion bars shall be selected at the rate of 1 per each mill heat or 1 of each 400 torsion bars produced, whichever occurs first.

4.5.2.2 Noncompliance. If a sample fails to pass Group D inspection, the contractor shall notify the Government and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted and on all units of product which are manufactured under essentially the same materials and processes, and which are considered subject to the same failure. After the corrective action has been taken, Group D inspection shall be repeated on additional sample units (or the reworked failed unit(s) if so identified and agreed upon in the corrective action plan submitted by the contractor). All tests and examinations, or the test(s)/examination(s) which the original sample failed, shall be performed at the option of the Government. Acceptance and shipment of product shall be discontinued until corrective action, acceptable to the Government, has been taken and Group D inspection has shown that the corrective action was successful.

4.5.2.3 Disposition of Group D samples. Samples torsion bars which have been subjected to group D inspection shall not be delivered as new equipment under the contract. Torsion bars subjected to group D inspection shall be indelibly marked "DO NOT USE".

TABLE V. Group D inspection.

Inspection	Requirement	Inspection method
Endurance	3.2.5	4.6.5

4.5.3 Inspection of packaging. The sampling and inspection of packaging shall be in accordance with the quality assurance provisions of the applicable packaging standard or packaging data sheet specified by the Government.

4.6 Methods of inspection.

4.6.1 Heat treatment. Following final heat treatment, the sample torsion bar or control specimen (see 4.6.1.1) shall be sectioned using a cut-off machine that will not change the structure by overheating during sectioning. The section shall be ground and polished using proper metallographic procedure and examined for conformance to 3.2.2.1 and 3.2.2.1.1.

4.6.1.1 Control specimen. If a control specimen is used, it shall meet the following criteria:

a. Same diameter as the torsion bar being heat treated.
b. 6 inches minimum length.

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c. Fabricated from material of the same heat as the torsion bars being sampled.
d. Have gone through the same heat treat process at the same time.
e. Marked with the raw material heat identification number at one end.

4.6.2 Surface indications. To determine conformance with 3.2.2.2, after final heat treatment and before any cold working, longitudinal and circular magnetic particle inspection shall be performed in accordance with ASTM E1444 using the fluorescent wet continuous method. The torsion bars shall be demagnetized following magnetic particle inspection.

4.6.3 Surface finish. Prior to cold working and application of protective coating, using comparator blocks, the surface finish shall be visually inspected for conformance to 3.2.2.2, 3.2.2.3 and 3.4.

4.6.4 Preset. The presetting operations shall be observed to verify that the requirements of 3.2.2.4 are met and that the preset direction arrow, as applied to the torsion bar, is compatible with the direction of windup.

4.6.5 Endurance. To determine conformance with 3.2.5, the torsion bar under test shall be installed in a suitable testing machine and deflected as specified in 3.2.5 until failure or 200,000 cycles maximum under continuous load with automatic or manual compensation for angular creep after 5000 cycles and every 5000 cycles thereafter. Torsional deflection and angular creep shall be as specified in 3.2.5 for the torsion bar under test. Rate of loading shall be between 10 and 100 cycles per minute and the temperature of the test specimen shall not exceed 250 °F. Anchors used in the testing machine shall be of the same spline configuration and hardness as those specified for use in operation of the vehicle. For group D inspection, the torsion bar shall be cycled for a minimum of 100,000 cycles (see 4.5.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. Torsion bars manufactured in accordance with this specification are intended for use on military tracked vehicles in the Bradley Fighting Vehicle (BFV) Family. This torsion bar is considered military unique as there are no commercially available counterparts.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, revision, and date of this specification.
b. Title, number, revision, and date of part drawing.
c. If required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
d. If first article inspection is required (see 3.1).
e. If a billet material sample is required (4.2.1).
f. If inspection conditions should be other than as specified (see 4.3).
g. Sampling plan requirements for group B inspection (see 4.5.1.3.1).
h. Torsion bars may be accepted for delivery to prime contractor at their risk prior to completion of control testing. However, components may not be presented for final Government acceptance without control tests completed (see 4.5.2).
i. If group D sampling should be other than as specified (see 4.5.2.1).
j. Packaging requirements (see 5.1).

6.3 First article. The acquisition document should include specific instructions regarding the arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive first article inspection, or any portion thereof, to those bidders offering a product which has been previously acquired or tested by the Government. Bidders offering such products should furnish evidence with the bid that prior first article approval is presently appropriate for the pending contract.

6.4 Recommended group B sampling plan (see 4.5.1.3.1).

6.4.1 Basic method. Quality characteristics listed in table III should be inspected using the sampling plan shown in table VI. Alternate sampling plans are not allowed except under an approved statistical process control (SPC) system conforming to the requirements outlined in 6.4.2.

6.4.1.1 Tightened inspection. Tightened inspection should be introduced as soon as 2 out of 5 successive lots have been rejected and should, as a minimum, impose a 30 percent increase in sample size. Normal inspection sampling may be restored after 5 successive lots have been accepted under tightened inspection.

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TABLE VI. Group B sampling plan.

Lot Size	Sample Size	
	For Major Characteristics	For Minor Characteristics
2 to 13	Entire lot	3
14 to 25	13	3
26 to 50	13	5
51 to 90	13	6
91 to 150	13	7
151 to 280	20	10
281 to 500	29	11
501 to 1200	34	15
1201 to 3200	42	18
3201 to 10 000	50	22

NOTE: Accept lot on zero defects.
Reject lot on one or more defects.

6.4.1.2 Reduced inspection. Reduced inspection may be introduced when 10 successive lots have been accepted and should as a maximum, permit a 30 percent decrease in sample size. Normal inspection sampling should be restored if a lot is not accepted under reduced inspection.

6.4.2 Statistical process control (SPC). Sampling plan requirements may be satisfied through the use of SPC as an alternate control method. If SPC is to be used, an SPC plan, detailing the methods for monitoring process control, should be approved by the Government or agent. The plan, developed by the contractor, should include as a minimum:

a. Types of control charts and their uses.
b. Process capability (C_{pk}) studies.
c. Criteria for determining out-of-control conditions.
d. Corrective actions to be taken if an out-of-control or out-of-tolerance condition is detected.
e. Training programs and qualification requirements of personnel executing the plan.
f. Provisions for reduced inspection.

The plan should contain the results of process capability studies for the processes being controlled. For variable data, the minimum C_{pk} for statistical control should be 1.33. For attribute data, a minimum process average of 99.73% is required.

6.5 Shot peening. In order to meet the endurance requirements of 3.2.5, previous manufacturing processing has included shot peening in accordance with SAE AMS 2430 as follows:

a. The torsion bar was shot peened before and after preset except that the intensity was 0.007C to 0.010C on the body and not less than 0.012A on the splines.

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b. Shot was round, cast steel with a hardness of Rockwell C55 through C65. Shot size on the torsion bar body was optional as required to obtain the specified intensity. Shot size used in the spline root area was not greater than 170 as defined in SAE AMS 2430.
c. Minimum coverage was 200 percent. Complete coverage of the spline root area was required.
d. All bar straightening and removal of surface indications was to be accomplished prior to peening. In addition, inspection of shot peening results were included in this document as follows:

    1. A minimum of one shot peening test strip was selected per each hour of processing for shot peening unless an approved SPC plan was used.
    2. Results of shot peening were included as group C inspections in table IV. Shot peen coverage, both before and after preset, was inspected using either a 10X magnifier or dyescan tracer liquid. Shot peen intensity was inspected using a test strip and by visual observation.

6.6 Protective covering. In order to protect the bar body from scratches and nicks, past manufacturing techniques have included the use of a tape covering. The body of the bar was covered with tape conforming to type EF (9 mm thick) of A-A-59770 in the area shown on Drawing 12292010. Tape was spirally wound with 50 percent overlap. Free ends had an additional winding to prevent peelback.

6.7 Definitions.

6.7.1 Torsion bar. The torsion bar is a round steel bar that acts as a torsional spring for the road arms of an armored, tracked vehicle.

6.8 Subject term (key word) listing.

Bar stock
Billet
Heat treatment
Involute tapered tooth
Macroetch
Master plug
Ring gage
Shot peen
Spline gage
Ultrasonic inspection

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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